

Algebra I Readiness Review for Algebra II Honors

Instructions

Complete in 45 minutes. Show clear algebraic work and check solutions when possible. Give exact answers using fractions and radicals; do not use rounded decimals. This review covers foundational skills expected at the beginning of Algebra II Honors.

Quick Reference

$$m = \frac{y_2 - y_1}{x_2 - x_1}, \quad y - y_1 = m(x - x_1), \quad y = mx + b$$

$$a^m a^n = a^{m+n}, \quad \frac{a^m}{a^n} = a^{m-n}, \quad (a^m)^n = a^{mn}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}, \quad (a + b)^2 = a^2 + 2ab + b^2, \quad a^2 - b^2 = (a - b)(a + b)$$

Problems

1. Simplify completely:

$$4(2x - 3) - 3(x + 5) + 2(1 - 2x).$$

2. Solve and check:

$$\frac{3}{4}(2x - 8) - \frac{1}{3}(x + 6) = 5.$$

3. Solve each inequality. Write each answer in interval notation.

(a) $5 - 3(2x - 1) > 4x + 8$

(b) $|2x - 5| \leq 9$

4. The points $(-2, 5)$ and $(4, -1)$ lie on line ℓ .

(a) Find the slope of ℓ and write its equation in slope-intercept form.

(b) Write the equation of the line parallel to ℓ through $(3, -4)$.

5. Let $f(x) = x^2 - 4x + 1$.

- (a) Find $f(-3)$.
- (b) Solve $f(x) = 1$.
- (c) Find the average rate of change of f from $x = 1$ to $x = 5$.

6. Solve the system exactly:

$$\begin{cases} 2x + 3y = 7, \\ 4x - y = 5. \end{cases}$$

7. Simplify completely. Use positive exponents and rationalize denominators.

(a) $\left(\frac{6x^{-2}y^3}{3xy^{-1}}\right)^2$

(b) $\sqrt{72} - 2\sqrt{8} + \sqrt{18}$

(c) $\frac{5}{2 - \sqrt{3}}$

8. Perform the indicated operations and simplify.

(a) $(2x - 3)(x^2 + x - 4)$

(b) Divide $2x^3 - 3x^2 - 11x + 6$ by $x - 3$.

9. Factor completely over the integers.

(a) $2x^2 - 7x - 15$

(b) $x^3 - 4x^2 - x + 4$

(c) $9y^2 - 25$

10. Solve exactly.

(a) $2x^2 - 3x - 5 = 0$

(b) $x^2 - 6x + 2 = 0$

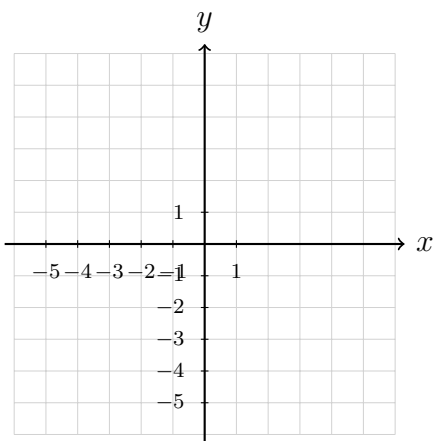
(c) $3x^2 = 27$

11. For $y = -x^2 + 4x + 5$:

(a) Find the vertex, axis of symmetry, and intercepts.

(b) State the domain and range.

(c) Graph the function on the coordinate plane.



12. A rectangle has an area of 96 square feet. Its length is 4 feet greater than its width.
- Write a quadratic equation that models the situation.
 - Find the rectangle's dimensions.
 - If both dimensions are increased by 25%, by what percent does the area increase?